

Ch8. Graphing Polynomials

1. The **End Behavior of the polynomials** and the **Leading Coefficient Test**:

As x goes to ∞ or $-\infty$, the graph of polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x^1 + a_0, \quad (a_n \neq 0)$$

either **risers** or **falls** eventually. Here, we can conclude this into the following table

degree n is an odd number		n is an even number	
$a_n > 0$	$a_n < 0$	$a_n > 0$	$a_n < 0$

2. Roots of a Function and x -intercepts.

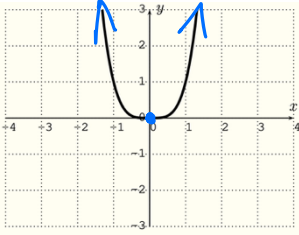
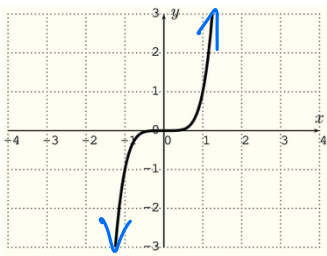
A root, or zero, or x -intercept of a polynomial $f(x)$ is a **number c** so that $f(c) = 0$.

Each **real** root/zero/solution of the polynomial $f(x)$ appears as an x -intercept of the graph of $f(x)$.
(Here 'real' means not a complex number)

Degree	Number of the roots	Graph of the polynomial of that degree		
2	At most 2 (real) roots 2 roots $\left\{ \begin{array}{l} 2 \text{ real roots} \\ 1 \text{ repeat root} \\ 2 \text{ complex roots} \end{array} \right.$			
3	At most 3 (real) roots			
4	At most 4 (real) roots			

3. Multiplicity of the root and x -intercepts.

Let $f(x) = (x - r)^n$ where r is the root of f and this root repeats n times. We call r a root with multiplicity n .

Even Multiplicity (n is even)	Odd Multiplicity (n is odd)
 $y = x^n$, for n even The graph <u>touches</u> the x-axis and <u>turns around</u> at the root r.	 $y = x^n$, for n odd The graph <u>crosses</u> the x-axis at the root r.
The graph tends to flatten out near the roots with multiplicity greater than <u>1</u>	

4. Turning Points of Polynomial Functions:

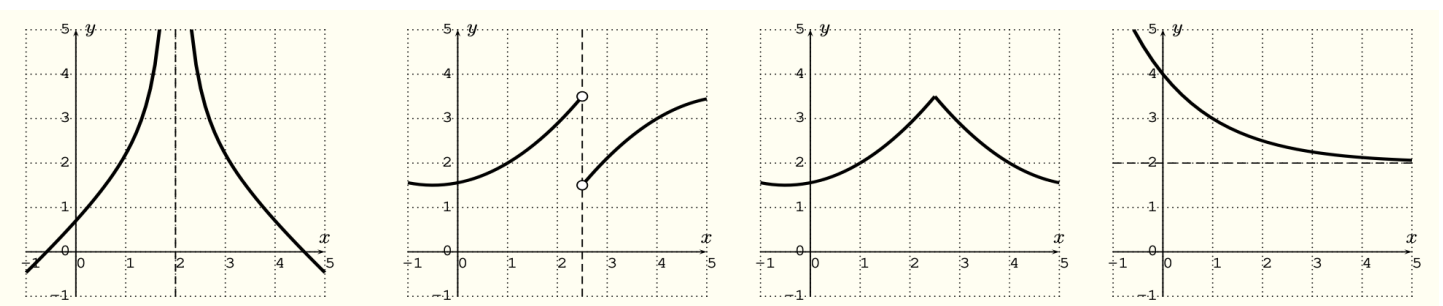
Let $f(x)$ be a polynomial function of **degree** n , then the graph of f has at most $n-1$ turning points.

5. The essential part for drawing a complete graph of f :

- End Behavior by leading coefficient test (how the function behaves when x approaches $+\infty$ or $-\infty$)
- All roots (which are x -intercepts) with the Multiplicities
- All y -intercepts (the values by computing $f(0)$ when $x=0$)
- All asymptotes (for rational functions in next chapter)
- Turning points with Extrema (that is all maximum and minimum)

6. The domain of a polynomial f is All Real number, and it is continuous for all real numbers and there are no jumps, no vertical or horizontal asymptotes, and no corner/cusp

The following graphs **cannot** be graphs of polynomials:



Vertical
asymptote

jumps

corner/cusp

horizontal
asymptote